Respiratory Disease Smart Assistant

This project focuses on developing an AI-powered Respiratory Disease Smart Assistant that can predict and detect respiratory diseases using symptom-based prediction and computer vision techniques. The system uses Flask for the web interface, Machine Learning for symptom prediction, and YOLOv8 for image-based disease detection.

1. Problem Statement

Respiratory diseases such as Asthma, Pneumonia, and Respiratory Infections are becoming increasingly common. Early diagnosis is important, but many people do not have immediate access to healthcare facilities. Traditional diagnosis methods require expert analysis and can take significant time. There is a need for an intelligent system capable of assisting users by predicting respiratory diseases based on symptoms and medical image analysis.

2. Existing Solution

Existing systems mainly rely on manual diagnosis by doctors or basic symptom checkers. Most systems either use only symptom-based analysis or only image classification methods. These systems often lack: Real-time disease prediction Integrated AI-based image analysis User-friendly interfaces Multi-disease detection capability Automated annotation and object detection

3. Proposed Solution

The proposed Respiratory Disease Smart Assistant combines symptom-based prediction with YOLOv8 image detection technology. The system predicts diseases such as: Asthma Pneumonia Respiratory Infection Normal Condition The application provides: Web-based user interface using Flask Machine Learning prediction using symptoms YOLOv8-based image detection Dataset annotation support Fast and intelligent disease prediction

4. Solution Workflow

Step	Description
1	Collect respiratory disease videos and images
2	Convert videos into image frames
3	Annotate images using LabelMe
4	Convert JSON annotations into YOLO TXT format
5	Split dataset into train and validation folders
6	Train YOLOv8 model using annotated dataset
7	Save trained model (best.pt)
8	Test model on unseen respiratory images
9	Integrate trained model into Flask web application
10	Display prediction results to users

5. Technologies Used

Python Flask YOLOv8 OpenCV Machine Learning Pandas Scikit-learn LabelMe Annotation Tool

6. Advantages

Fast respiratory disease prediction Automated image analysis User-friendly interface Supports multiple disease classes Low-cost AI-based healthcare assistant Can assist early disease identification

7. Future Scope

Future improvements can include: Real-time webcam disease detection Cloud deployment Mobile application integration Voice-based symptom assistant Improved deep learning accuracy with larger datasets

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Annotation Link:
https://drive.google.com/drive/folders/1ZYBcSwiOUX416Y1--E6xtg8ciZrb7r6B?usp=drive_link

8. Conclusion

The Respiratory Disease Smart Assistant provides an intelligent and automated approach for detecting respiratory diseases using Artificial Intelligence and Deep Learning. The system combines symptom prediction and image detection to assist users with early-stage respiratory disease analysis. This project demonstrates how AI can improve healthcare accessibility and support smart medical diagnosis systems.